

Detailed FEM Calculation Flow

assemble_global_matrix()

- `n = nodes.size()` (216)
- `global_K[n×n] = 0`
- `global_F[n] = 0`

For each element `e` (0~749)

- `nodes[4]: n0, n1, n2, n3`
- `σ` (material property)
- Jacobian `J[3×3]`
- `detJ = Jacobian`
- `volume = detJ / 6`

For each Gauss (4)

- (Gauss point coordinates)
- `B = J^-1 * dN/dξ`
- `Ke += σ * B^T * B * detJ * weight`

- `global_K[gi][gj] += Ke[i][j]`
- `global_F[gi] += Fi`

solve_linear_system() - Gauss-Seidel Method

- `potentials[]` (initial guess)
- `max_iter = nstep = 100`
- `while max_diff > 1e-6`

(iter loop)

- `max_diff = 0`
- For each node `i` (0~215):
 - `fixed_node[i]` (boundary condition)
 - `global_K[i][i] == 0`
 - `sum = global_F[i]`
 - `sum -= Σ(global_K[i][j] * potentials[j]) (i≠j)`
 - `new_phi = sum / global_K[i][i]`
 - `diff = |new_phi - potentials[i]|`
 - `potentials[i] = new_phi`

- `max_diff < tol`
- `max_diff`
- (return values)